MDTS4214 PROBLEM SET 7

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Comparison of least squares, ridge regression, LASSO and Elastic Net

```
rm(list=ls())
library(ISLR)
## Warning: package 'ISLR' was built under R version 4.3.3
df=na.omit(Hitters) #1. Removing the missing rows
dim(df)
## [1] 263  20
#2. Splitting into Train and Test
set.seed(123)
test.id=sample(1:263,100)
train.data=df[-test.id,] #Training data
head(train.data)
##                  AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun
## -Alan Ashby        315   81     7   24  38    39    14   3449   835     69
## -Alvin Davis       479  130    18   66  72    76     3   1624   457     63
## -Andre Dawson      496  141    20   65  78    37    11   5628  1575    225
## -Alfredo Griffin   594  169     4   74  51    35    11   4408  1133     19
## -Andres Thomas     323   81     6   26  32     8     2    341    86      6
## -Andre Thornton    401   92    17   49  66    65    13   5206  1332    253
##                  CRuns CRBI CWalks League Division PutOuts Assists Errors
## -Alan Ashby        321  414    375      N        W     632      43     10
## -Alvin Davis       224  266    263      A        W     880      82     14
## -Andre Dawson      828  838    354      N        E     200      11      3
## -Alfredo Griffin   501  336    194      A        W     282     421     25
## -Andres Thomas      32   34      8      N        W     143     290     19
## -Andre Thornton    784  890    866      A        E       0       0      0
##                  Salary NewLeague
## -Alan Ashby         475         N
## -Alvin Davis        480         A
## -Andre Dawson       500         N
```

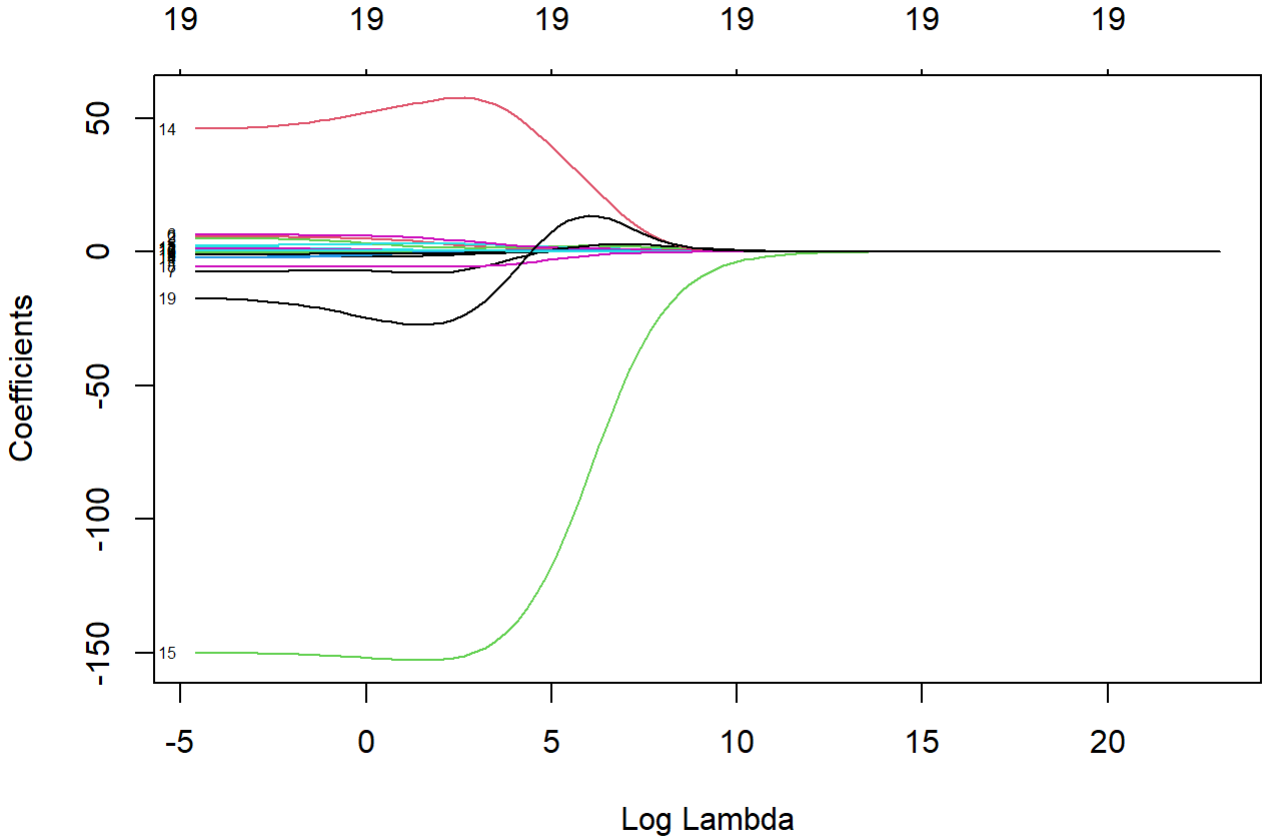
```
## -Alfredo Griffin    750         A
## -Andres Thomas       75         N
## -Andre Thornton    1100         A
test.data=df[test.id,] #Test data
head(test.data)
##                AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns
## -Marvell Wynne   288   76     7   34  37    15     4   1644   408     16   198
## -Bill Almon      196   43     7   29  27    30    13   3231   825     36   376
## -Rick Cerone     216   56     4   22  18    15    12   2796   665     43   266
## -Jack Howell     151   41     4   26  21    19     2    288    68      9    45
## -Spike Owen      528  122     1   67  45    51     4   1716   403     12   211
## -Tom Paciorek    213   61     4   17  22     3    17   4061  1145     83   488
##                CRBI CWalks League Division PutOuts Assists Errors Salary
## -Marvell Wynne  120    113      N        W     203       3      3    240
## -Bill Almon     290    238      N        E      80      45      8    240
## -Rick Cerone    304    198      A        E     391      44      4    250
## -Jack Howell     39     35      A        W      28      56      2     95
## -Spike Owen     146    155      A        W     209     372     17    350
## -Tom Paciorek   491    244      A        W     178      45      4    235
##                NewLeague
## -Marvell Wynne         N
## -Bill Almon            N
## -Rick Cerone           A
## -Jack Howell           A
## -Spike Owen            A
## -Tom Paciorek          A
dim(train.data);dim(test.data)
## [1] 163  20
## [1] 100  20
#Train the model using least squares and report the train MSE and test MSE.
m1=lm(Salary~.,data=train.data)
mse.train=1/163*(sum((train.data$Salary-predict(m1))^2))
y.hat.test=predict(m1,newdata=test.data)
mse.test=mean((test.data$Salary-y.hat.test)^2)
mse.train
## [1] 105334.2
mse.test
## [1] 87803.91
#3. Fit Ridge Regression
```

```
library(glmnet)
## Warning: package 'glmnet' was built under R version 4.3.3
## Loading required package: Matrix
## Loaded glmnet 4.1-8
lambda.grid = 10^seq(10, -2, length = 100)
x=model.matrix(Salary~.,train.data)[,-1]
x.test=model.matrix(Salary~.,test.data)[,-1]
head(x)
##                   AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun
## -Alan Ashby         315   81     7   24  38    39    14   3449   835     69
## -Alvin Davis        479  130    18   66  72    76     3   1624   457     63
## -Andre Dawson       496  141    20   65  78    37    11   5628  1575    225
## -Alfredo Griffin    594  169     4   74  51    35    11   4408  1133     19
## -Andres Thomas      323   81     6   26  32     8     2    341    86      6
## -Andre Thornton     401   92    17   49  66    65    13   5206  1332    253
##                   CRuns CRBI CWalks LeagueN DivisionW PutOuts Assists Errors
## -Alan Ashby         321  414    375       1         1     632      43     10
## -Alvin Davis        224  266    263       0         1     880      82     14
## -Andre Dawson       828  838    354       1         0     200      11      3
## -Alfredo Griffin    501  336    194       0         1     282     421     25
## -Andres Thomas       32   34      8       1         1     143     290     19
## -Andre Thornton     784  890    866       0         0       0       0      0
##                   NewLeagueN
## -Alan Ashby                1
## -Alvin Davis               0
## -Andre Dawson              1
## -Alfredo Griffin           0
## -Andres Thomas             1
## -Andre Thornton            0
ridge.model=glmnet(x, train.data$Salary, alpha = 0,lambda=lambda.grid)
dim(coef(ridge.model)) # (i) Dimension of coefficient matrix
## [1]  20 100
beta50 = coef(ridge.model)[,50]
beta60 = coef(ridge.model)[,60]

l2.50 = sqrt(sum(beta50^2))
l2.60 = sqrt(sum(beta60^2))

l2.50
```

```
## [1] 438.5671
l2.60 # (ii) L2 Norms
## [1] 124.7542
# Interpretation :
# As λ increases → coefficients shrink → L2 norm decreases
# So, l2.60 < l2.50

pred50 = predict(ridge.model, s = lambda.grid[50], newx = x.test)
pred60 = predict(ridge.model, s = lambda.grid[60], newx = x.test)

mse50 = mean((test.data$Salary - pred50)^2)
mse60 = mean((test.data$Salary - pred60)^2)

mse50
## [1] 167303.1
mse60 # (iii) Test MSE for λ
## [1] 97935.56
```

4. Plot of Coefficients vs λ

```
plot(ridge.model, xvar="lambda", label=TRUE)
```



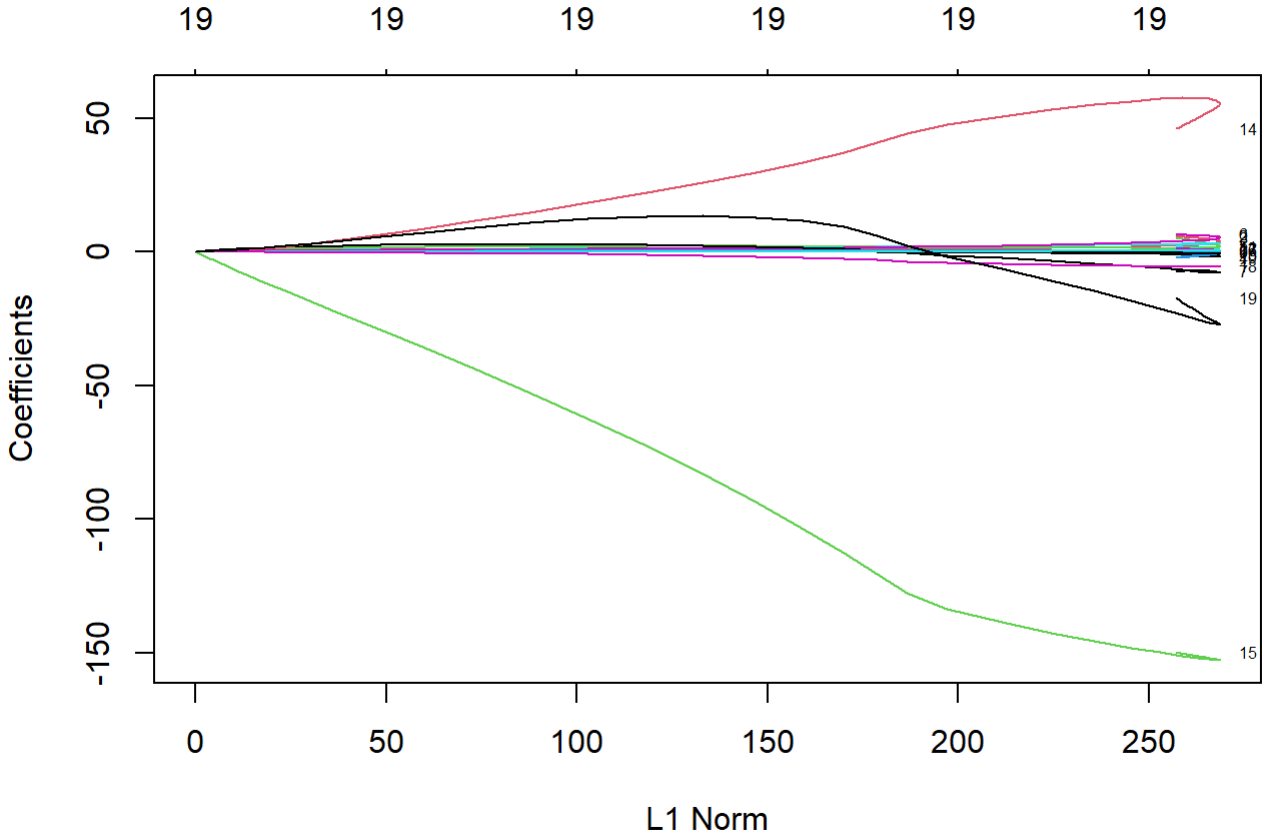
Interpretation: As λ increases → coefficients shrink toward 0.

Some shrink faster than others.

No coefficient becomes exactly zero.

5. Plot vs L2 Norm Ratio

```
plot(ridge.model, xvar="norm", label=TRUE)
```



Interpretation :

The graph becomes more intuitive than λ.

It Starts from: 1 → least squares

towards 0 → heavy shrinkage

Shape remains simlar, just rescaled.

6.Ridge with cross validation

```
set.seed(123)
cv.ridge=cv.glmnet(x, train.data$Salary, alpha=0)

best.lambda=cv.ridge$lambda.min
best.lambda
## [1] 404.3208
pred=cv.predict=cv.predict <- predict(cv.ridge, s=best.lambda, newx=x.test)

mse.cv=mean((test.data$Salary - pred)^2)
mse.cv
## [1] 93690.09
```

7. LASSO Regression

```
set.seed(123)
cv.lasso=cv.glmnet(x, train.data$Salary, alpha=1)

best.lambda.lasso=cv.lasso$lambda.min

lasso.pred=predict(cv.lasso, s=best.lambda.lasso, newx=x.test)

mse.lasso=mean((test.data$Salary - lasso.pred)^2)
mse.lasso
## [1] 93914.89
coef.lasso=coef(cv.lasso, s=best.lambda.lasso)

sum(coef.lasso!=0) # No. of non-zero coefficients.
## [1] 9
```

8. Elastic Net

```
set.seed(123)
cv.enet=cv.glmnet(x, train.data$Salary, alpha=0.5)

best.lambda.enet=cv.enet$lambda.min

enet.pred=predict(cv.enet, s=best.lambda.enet, newx=x.test)

mse.enet=mean((test.data$Salary - enet.pred)^2)
mse.enet
## [1] 94149.71
coef.enet=coef(cv.enet, s=best.lambda.enet)
coef.enet[coef.enet!=0] # Non-zero Coefficients
## [1]  107.38690040    0.56044286    2.34354343    1.34246607    0.09246948
## [6]    0.12317205    0.20485216 -111.90806113    0.28419816
sum(coef.enet!=0)
## [1] 9
```

9. Comparison of Key features of the Test mse of different Regression techniques:

For, Least Squares → No regularization

Ridge → Shrinkage

LASSO → Variable selection

Elastic Net → Hybrid

Conclusion:

Ridge reduces variance keeping all predictors.

LASSO removes irrelevant predictors.

Elastic Net combines both advantages of Ridge and LASSO.

Cross-validation gives optimal λ.

So,Regularized models predict the Salary of a baseball player more accurately than least squares.